BACK-EMF TESTER
HMI Engineering Handover Specification

Document Version: 1.0
Status: Released for HMI Development
PLC Platform: ABB AC500-eCo PM5032-T-ETH
Communication: Modbus TCP
Prepared By: PLC Controls Engineering

1. Purpose

This document defines the complete communication interface between the ABB PLC and the HMI. It includes:

- Communication settings
- Register map
- Machine states
- Alarm definitions
- User interface rules
- Safety requirements
- Data logging requirements

No additional PLC tags are required for HMI development.

2. Communication

Parameter	Value
Protocol	Modbus TCP
PLC Role	Server (Slave)
HMI Role	Client (Master)
PLC IP	192.168.0.1
Port	502
Register Type	Holding Registers (4xxxx)
Data Type	UINT16

Recommended polling interval:

- Critical Status: 100 ms
- Process Values: 200 ms
- Logging Buffer: Read only after test completion

3. Register Map

3.1 HMI → PLC (Write)

Register	PLC Variable	Description	Range
40001	uiSpeedRPM	Target Test Speed	1–18000 RPM
40002	uiAccelTime	Acceleration Time	1–60 sec
40003	uiDecelTime	Deceleration Time	1–60 sec
40004	uiHeartbeat	Heartbeat Counter	Increment continuously
40005	uiTestTimeSec	Test Duration	1–600 sec

3.2 PLC → HMI (Read)

Machine Information

Register	PLC Variable	Scaling	Description
40020	uiCyclesLow	None	Production Counter Low Word
40021	uiCyclesHigh	None	Production Counter High Word

Live Process Data

Register	PLC Variable	Scaling	Description
40030	uiActSpeedRPM	None	Measured Shaft RPM
40031	uiVfdActualRpm	None	VFD Reported RPM
40032	uiLiveVoltageUV_x100	÷100	Live UV Voltage
40033	uiLiveVoltageVW_x100	÷100	Live VW Voltage
40034	uiLiveVoltageWU_x100	÷100	Live WU Voltage
40035	uiPeakVoltageUV_x100	÷100	Peak UV Voltage
40036	uiPeakVoltageVW_x100	÷100	Peak VW Voltage
40037	uiPeakVoltageWU_x100	÷100	Peak WU Voltage

Machine Status

Register	PLC Variable	Description
40047	uiState	Machine State
40048	uiFault	Fault Code
40049	uiHealthy	Machine Healthy

40050	uiHMIFault	Heartbeat Fault

4. Scaling

Only voltage values require scaling.
Displayed Voltage
Displayed Value = Register Value / 100
Example
Register = 1543
Display = 15.43 V
RPM values are displayed directly.

5. Machine State Definitions

Value	Display
0	Initializing
1	Ready
2	Clamping Motor
3	Extending Sled
4	Waiting For VFD
5	Testing
6	Retracting Sled
7	Unclamping Motor
8	Fault

State 8 shall force the HMI into Alarm Mode.

6. Fault Codes

Code	Fault
0	No Fault
1	Emergency Stop
2	Air Pressure Lost
3	Safety Door Open
4	Clamp Timeout
5	Clamp Position Lost
6	Sled Extend Timeout
7	Sled Position Lost
8	Sled Retract Timeout
9	Cylinder Sensor Conflict
10	Actuator Command Conflict
11	Sequence Watchdog Timeout

Fault Banner shall always display the active fault.

7. Heartbeat Protocol

The HMI shall maintain communication using Register 40004.

Implementation: Increment this register once every second. (Example: 0, 1, 2, 3, 4, 5...)

If the PLC detects that the value has not changed for approximately five seconds:

- Machine stops
- Motion is safely terminated
- uiHMIFault becomes TRUE

Heartbeat is mandatory.

8. Machine Health

Register	Meaning
40049 = 1	Machine Ready
40049 = 0	Machine Fault

Display: Green = Healthy, Red = Fault

9. Operator Interface

Editable Parameters: Speed, Acceleration Time, Deceleration Time, Test Time.

Read Only: Actual RPM, VFD RPM, Live Voltages, Peak Voltages, Cycle Counter, Machine State, Machine Fault, Machine Health.

10. Operator Controls

The HMI SHALL NOT contain:

- Start Button
- Stop Button
- Reset Button

Machine operation is controlled using physical machine pushbuttons only.

11. Fault Screen

When: Machine State = 8

Display: FAULT ACTIVE
Display active alarm description.
Display: "Press the physical RESET pushbutton on the machine."

No software reset is permitted.

12. Trend Logging

The PLC stores one complete Back-EMF test.

- **Sample Count:** Register 40100
- **Valid Range:** 0–1000 Samples
- **Buffer Base Address:** 41000

Each sample occupies five words.

Offset	Description
0	Timestamp (0.1 sec)
1	RPM
2	UV Voltage ×100
3	VW Voltage ×100
4	WU Voltage ×100

Sample 0: 41000–41004
Sample 1: 41005–41009
Sample 2: 41010–41014
Continue until Sample Count.

13. Data Ready

Coil: 0x0200
When TRUE: Trend acquisition is complete.

The HMI shall:

- Read Sample Count
- Read the entire logging buffer
- Display the trend

14. Recommended Screens

- **Home:** Machine Status, Machine State, Fault Banner, Speed, Voltages, Cycle Counter, Healthy Indicator
- **Setup:** Target RPM, Acceleration, Deceleration, Test Time
- **Live Monitor:** Actual RPM, VFD RPM, UV, VW, WU
- **Trend Screen:** Back-EMF Graph (RPM, UV, VW, WU), Zoom, Pan, Export
- **Alarm Screen:** Current Fault, Fault History, Machine Status, Communication Status

15. Communication Failure

If communication is lost:

- Display "PLC Communication Lost"
- Disable parameter editing.
- Do not clear operator values.
- Reconnect automatically.

16. Polling Recommendations

Item	Poll Time
Machine Status	100 ms
Fault	100 ms
RPM	100 ms
Voltages	200 ms
Cycle Counter	500 ms
Trend Buffer	End of Test Only

17. Acceptance Criteria

The HMI shall be considered complete when:

- All registers read correctly
- Parameter writes update PLC
- Heartbeat functions continuously
- Fault banner updates automatically
- State indicator follows PLC sequence
- Trend graph displays logged data
- No Start or Reset controls exist
- Communication recovers automatically after cable reconnect

18. Register Summary

Register	Function
40001	Target Speed
40002	Acceleration
40003	Deceleration
40004	Heartbeat
40005	Test Time
40020	Cycle Counter Low
40021	Cycle Counter High
40030	Actual RPM
40031	VFD RPM
40032	Live UV
40033	Live VW
40034	Live WU
40035	Peak UV
40036	Peak VW
40037	Peak WU
40047	Machine State
40048	Fault Code
40049	Machine Healthy
40050	HMI Heartbeat Fault

40100	Logged Sample Count
41000+	Trend Buffer

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